Joshua Goldberg

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Education

Northeastern University - Boston, MA | Khoury College of Computer Sciences *Sept 2023 - Present*
Candidate for a B.S. in Computer Science, Minor in Mathematics *Expected Apr 2027*
Honors: GPA 4.0/4.0 | Dean's List (5/5 semesters) | Honors Scholarship
Relevant Courses: Software Engineering, Programming Languages, Algorithms (Graduate), Object Oriented Design

Knowledge/Expertise

Programming Languages: Lean, Java, Python, Javascript/Typescript, Rust, HTML/CSS, Rocq, OCaml, Racket, C
Applications/Systems: Git, Docker, Node.js, n8n, React, Postman, NumPy, Autograd, Prisma, Qt, IntelliJ, PyCharm

Projects

CatMode - *Lean 4* *Jan 2026 - July 2026*
- Worked with the Northeastern PRL to write a DSL supporting capabilities for reasoning using categorical structures
- Designed definitions to represent information flow and security lattices within the context of categories

OwlLean [source] - *Lean 4* *Aug 2025 - June 2026*
- Developed a type checker for Owl, integrating subtyping and label corruption across 14,000+ lines of code to guarantee safety of secret data from outside adversaries
- Implemented bidirectional typing inference and automated 10+ tactics for proof generation, ensuring proper type checking of information flow constraints across 5+ example cryptographic programs
- Devised performance optimizations, allowing runtimes to be over 6x faster compared to unoptimized tests

Agentic Cybersecurity Bot [source] - *Python* *June 2025 - Dec 2025*
- Led a team of 8 developers in the creation of a **Retrieval-Augmented Generation (RAG)** integrated Discord bot leveraging the **GPT-4 API**, and n8n workflow automation to assist 200+ students with cybersecurity coursework
- Authored and presented research at CISSE, demonstrating system design with an over 80% student helpfulness rating

Osu!mania Replay Renderer [source] - *Java* *July 2024 - Mar 2025*
- Implemented a MVC application that parses **LZMA** compressed game data and converts resulting data to MP4 format by recreating the game environment, using **7Zip** and **FFmpeg**
- Optimized object-oriented design patterns within the renderer, decreasing average render times by over 50%

Experience

Incoming Software Developer Intern - IBM *Aug 2026*

Software Engineer Intern, Global Supplier Tech - Wayfair *June 2026 - Aug 2026*
- Built a **Python MCP** server with 12+ tools for supplier LLM agents to query key data from 7+ advertisement endpoints, including campaign data, performance report generation, SKU eligibility, and internal recommendation services
- Implemented OAuth protocols for token isolation enabling server deployment for use amongst 100+ employees.
- Translated group assigners and data logging infrastructure into **Rust**, with estimations of over 2x faster program runs

Research Assistant - Northeastern Programming Research Laboratory *July 2025 - Present*
- Building automated tools in **Lean 4** and **Rocq** to verify security properties of cryptographic programming languages (e.g. Owl), enabling alternatives to manual proof goals
- Communicated novel implementation ideas and future plans during weekly sprints to ensure development success
- Spearheaded the formal verification of 4+ critical security properties across 3 distinct language representations

Technical Lead - Northeastern University Electric Racing *Apr 2025 - June 2026*
- Managed a team of 5+ developers using **React** and **Typescript**, assigning 50+ tickets and reviewing 30+ pull requests
- Built full-stack improvements for the Finance, Gantt Chart, and Calendar pages, integrating **Prisma** and **SQL filtering** and cutting management overhead and improving load times by over 50% for 200+ members across sub teams

Logic and Computation Teaching Assistant - Northeastern University - Boston, MA *Jan 2025 - May 2026*
- Mentored 100+ students weekly in preparation for proof-based exams in formal logic and property-based testing
- Maintained autograder infrastructure in **Racket** and **OCaml**, designing systems to handle 100+ submissions weekly

Interests : Piano | Theorem Provers | Programming Languages | Formal Security | American Numismatics